



OPTICAL DRIVES

Whether it is a CD-ROM or a DVD drive, optical storage is the choice for data portability. But it can bog down your system if not configured right

When it comes to supplemental data storage, nothing has quite revolutionized portability like optical storage drives. Whether it is the CD-ROM, the DVD or the CD-RW, the convenience, portability of the media and comparatively low costs have been responsible for the immense success of optical storage devices. While the underlying technology behind these drives is primarily the same, where the surface of the CD media is coated with special dyes and lasers are used to read and write information to it, there is little else in terms of technology that differentiates these drives from one another.

Also, the performance increases of these peripheral devices has not been as dramatic as that of graphics cards or processors. However, with most of today's software and games being made available on CD-ROM, it is important that your optical drive is configured to its best working level.

If you've ever loaded an application such as Microsoft Office, you will notice that the process of reading files from the CD-ROM drive sometimes takes ages—you could easily take a 20-minute coffee break! Like your hard disk drive, CD-ROM drives also operate in one of two modes—Programmed Input Output (PIO) or Direct Memory Access (DMA). The latter is more efficient and speedy and therefore, if your CD/DVD drive is not set to DMA within Windows, there will be a loss in performance while installing large applications, watching VCDs or DVDs, and while playing games.

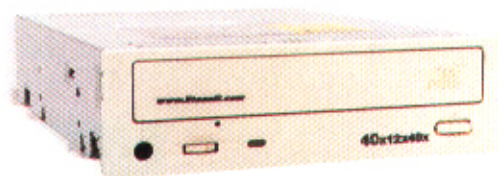
A very common misconception with CD and DVD drives is the correlation most users draw between the 'x' factor and the drive speed. In

most CD and DVD-ROM drives, a higher 'x' factor only represents the maximum data transfer rate your drive is theoretically capable of. In real world terms, even some of the fastest CD and DVD-ROM drives exhibit data transfer speeds approaching 7 MBps. With the vast majority of drives, a data transfer rate of 4 MBps is normal. With most optical drives today, a technique called Constant Angular Velocity is used. Here, the spindle of the drive spins at a constant rate throughout the operation, meaning that the data transfer speeds near the hub of the disk are lower than at the rim of the disk. Constant Linear Velocity (CLV) overcomes this situation by keeping the linear velocity constant along all areas of the disk and the drive's actual rotational speed is changed depending upon the location of the read head over the disk. These drives give uniform data transfer speeds during the course of their operation.

Did you find the playback choppy and jerky the last time you watched a VCD or DVD, even while using a powerful computer? The reasons for this could be manifold. If you've connected your CD or DVD drive as a slave to your hard disk, the VCD and DVD playback could be adversely affected each time data is simultaneously accessed from the hard disk since both devices are on the same IDE cable. It could also be that you have not enabled DMA for your drives. With DMA disabled, your DVD drive will call upon your main processor for every single data transfer request. Therefore, if your processor is already busy with decoding your DVD video stream, it would result in your DVD not being able

to pull out data fast enough, leading to a jerky playback.

Yet another factor that affects optical drives is a dirty laser pickup lens. Since they usually have retractable trays, there is an access point from where dust particles in the air can reach the innards of the drive, causing problems. The common



CD-Writers can take up a significant amount of system resources if they're not configured correctly

symptoms of this are poor speeds during CD-based applications, the inability to read media and sometimes even the rejection of a CD or DVD.

CD-RW drives expect to be fed with data in a constant flow in order to write a CD-R or CD-RW reliably. Now this is where problems can occur—have you ever been plagued with failed CD writes even though you've bought the fastest 48x CD-Writer? This could be attributed to an interruption in the flow of the data to the drive because the hard disk or processor is busy performing a processor-intensive action (such as playing MP3s) while a CD is being written. When the CD-RW drive fails to find the data in its buffer, the condition is known as a buffer underrun. It is safer to refrain from doing anything else on your PC while a CD is being written so as to ensure that there is no external disturbance during the write process.

Tips & Tweaks

- Make sure you enable DMA for your optical drives from the BIOS and within Windows.
- Enable any buffer underrun protection methodologies supported by your CD-RW drive during writes.
- Housekeeping chores on your optical drive such as running a pickup lens cleaner every couple of months, will help ward off problems in the long run.
- Keep a lookout for Flash updates to your drive's firmware—this can elicit additional functionality and also iron out any known bugs in your drive's firmware.



Though it is among the fastest optical drives available, in real world terms, DVD drives exhibit data transfer rates approaching 7 MBps